ADIDNS Time Bombs: Poison Today, Relay Tomorrow

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Predictable hostnames can be exploited with malicious ADIDNS records to take control of hosts added to the domain at a later date. This is made possible by the default permissions every domain user has over the creation of arbitrary DNS records and the way these records translate to `dnsNode` objects. I am calling these ADIDNS time bombs because the malicious records will sit in the directory indefinitely, granting the attacker full control of the target computer's A record as soon as it joins the domain.

Concept

I was reading Kevin Robertson's first article on ADIDNS when one small detail got the cogs spinning in my head: ADIDNS records are stored as `dnsNode` objects which get grouped by name rather than type. If you have three DNS records of the same name but of types A, MX, and NS, this results in a single `dnsNode` with three entries in its `dnsRecord` attribute.

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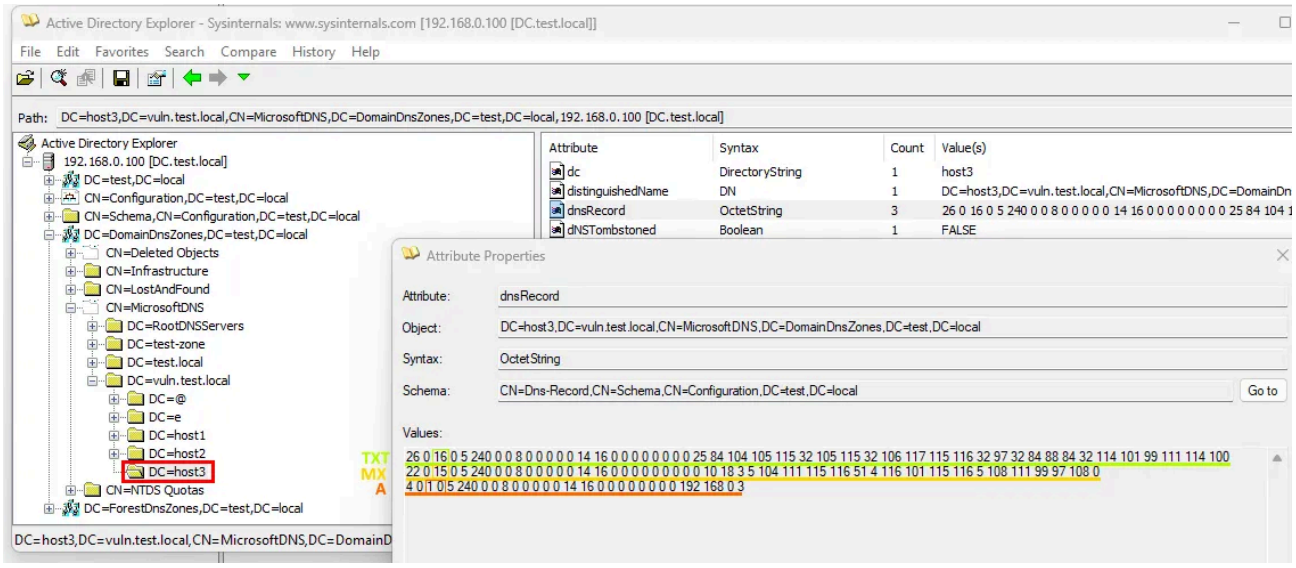

The owner of a `dnsNode` is usually its creator, and by default only the creator itself, domain and DNS admins have permissions to modify the node with all its records, as they all share the same DACL. The creator of a brand new `dnsNode` doesn't have full control, but almost: they can change the node's owner and DACL, as well as all the important attributes for objects of this kind.

These factors introduce the possibility of injecting malicious records with the name of hosts you think are going to be added to the domain in the near future: say for example you ran BloodHound and among the machine accounts you see names like these:

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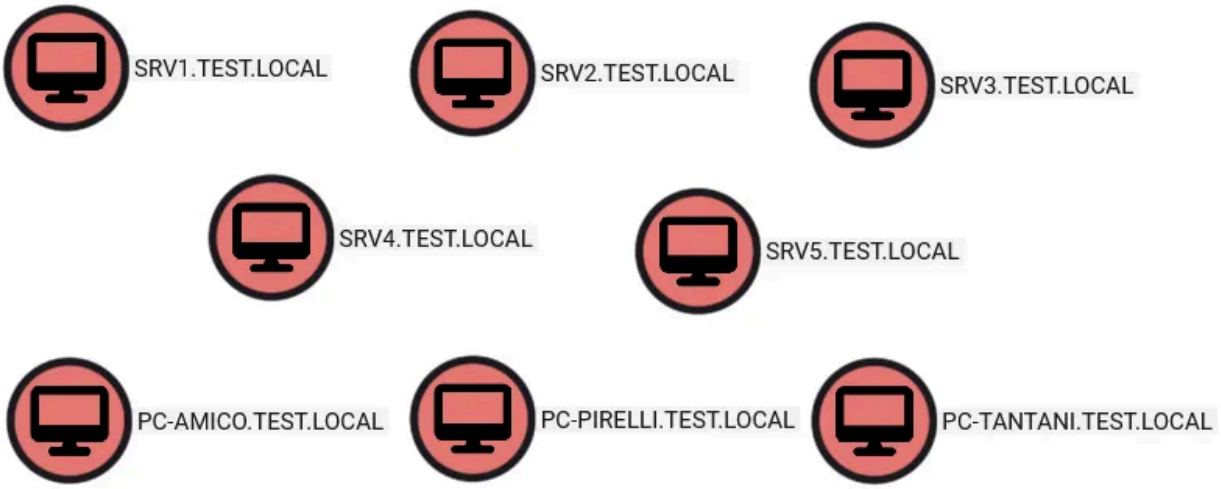

These names are predictable, incremental numbers, employee names, and so on. You could use this information to create one or more malicious ADIDNS records with names that follow this pattern, giving yourself full control over the A record of the future machines before they even exist, thus establishing a MITM position and taking over the computers with techniques like AP-REQ relays:

- identify predictable hostnames
- register one or more non-A records using the predictable names (eg: TXT)
- adjust `dnsNode` DACL to allow the computer to add its address
- wait for the target to join by monitoring the directory
- poison the A record with attacker's IP
- enjoy the kerberos relays!

The attack obviously requires time patience and some luck, but I thought it was a neat little trick to exploit the very mechanics of ADIDNS.

Pre-created machine accounts are another source of possible time bomb targets: look for recently added machine accounts without an A record and with no logon recorded in their `lastlogontimestamp` attribute, these might be computers that the administrators will be joining soon. This technique could also be effective in the hands of a disgruntled employee, anyone with access to the names of new staff before their computers are in the domain, or someone with knowledge of planned infrastructure additions, like a new web server to be deployed in the next few days.

Creating Malicious Records

On Linux we can combine the `krb5-user` kerberos client with nsupdate, a tool we find built onto many Linux distros which allows us to craft very specific DDNS requests. I have frequently shown dnstool.py from krbrelayx to create and modify DNS records, but as of

now only A records are supported and that's exactly the record we need to avoid for this attack.

After setting up `krb5-user` we request a TGT:

```
kinit amico@TEST.LOCAL
```

With `klist` we can double check the presence of a valid ticket. Now we can send an authenticated DDNS with `nsupdate`, this is possible thanks to its `gsstsig` command: to make it short, GSS-TSIG allows us to make authenticated DDNS (Dynamic DNS) updates over kerberos, so `gsstsig` picks up the TGT in our cache and uses it.

To make it long, if you're interested, GSS-TSIG (Generic Security Service Algorithm for Secret Key Transaction) is a protocol combining GSSAPI (Generic Security Service Application Programming Interface) and TKEY to protect DDNS with kerberos authentication. GSSAPI is an API used to add an authentication layer to other protocols, allowing developers to create secure protocols on top of existing authentication implementations. Another API you almost always see mentioned in conjunction with GSSAPI is SPNEGO (Simple and Protected GSSAPI Negotiation Mechanism), which is used to negotiate the authentication method to use with GSSAPI. Windows uses SPNEGO to determine whether to use NTLM or kerberos. Finally, TSIG (Transaction Signature) is the protocol DDNS uses for authenticated requests, such as ADIDNS "secure dynamic updates".

We also specify the target DNS server and zone, which usually will be the main one, but DNS zone enumeration should be performed to tell exactly where our predictable records are stored, because creating two records with the same name in different zones can have unpredictable results and could impact the host's availability. adidnsdump for example creates a CSV with every found record and can list DNS zones, as well as look for them in all three possible containers (`DomainDnsZones`, `ForestDnsZones`, or `System`).

With `update add` we provide record name, TTL, type and value. Any type other than A or AAAA should work, just make sure you provide a value that is valid for the selected type. For simplicity I use TXT. `send` submits the request.

```
gsstsig
server dc.test.local
zone test.local
update add srv6.test.local 86400 TXT abcdefghi
send
```

```
┌──(kali㉿kali)-[~]
└─$ kinit amico@TEST.LOCAL
Password for amico@TEST.LOCAL:

┌──(kali㉿kali)-[~]
└─$ klist
Ticket cache: FILE:/tmp/krb5cc_1000
Default principal: amico@TEST.LOCAL

Valid starting       Expires              Service principal
08/11/2025 07:49:32  08/11/2025 17:49:32  krbtgt/TEST.LOCAL@TEST.LOCAL
        renew until 08/12/2025 07:49:22

┌──(kali㉿kali)-[~]
└─$ nsupdate
> gsstsig
> server dc.test.local
> zone test.local
> update add srv6.test.local 86400 TXT abcdefghi
> send
>
```

As an alternative we can use a tool I'll talk more about later, godap. Bind to the DC, go to the ADIDNS tab, select the target zone and press `Ctrl + N` to add a record.

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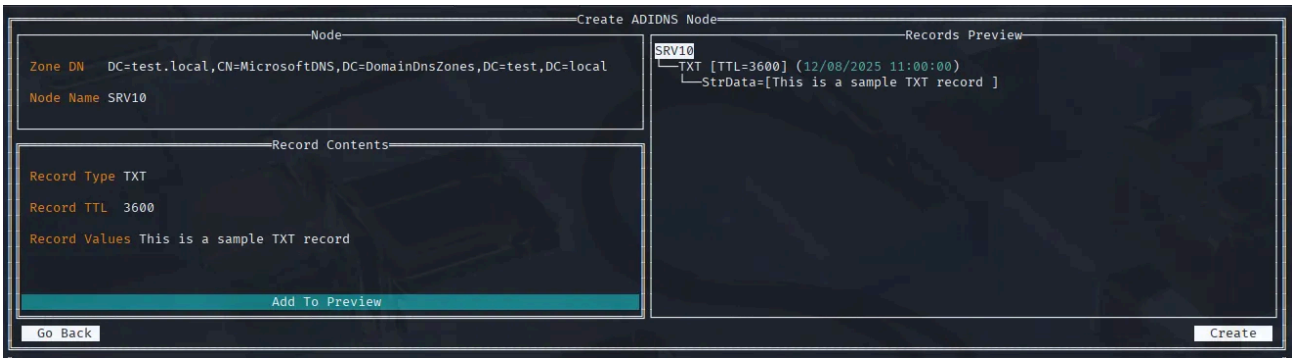

Tweaking The Node DACL

We can verify that the record was indeed created.

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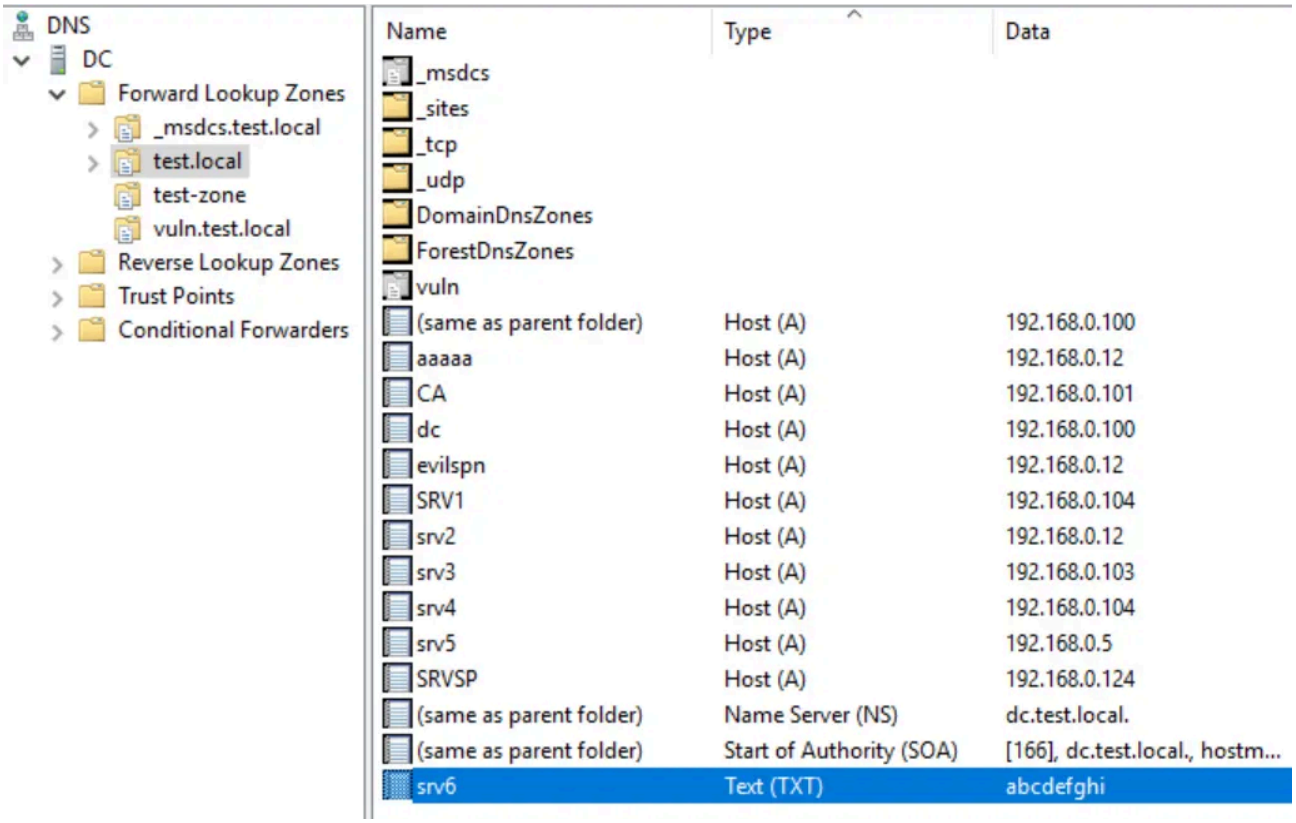
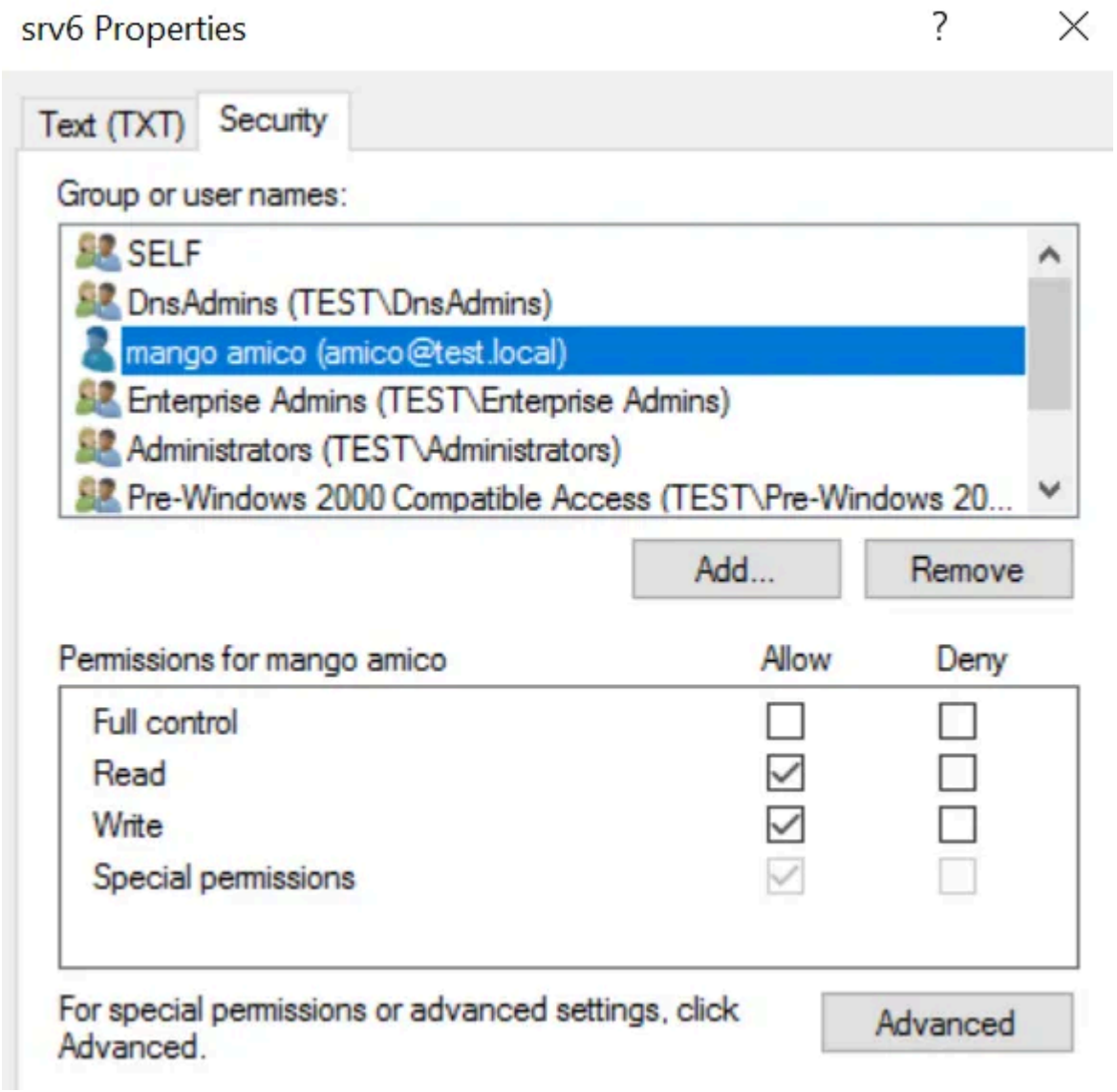


Our user, `amico`, is its owner and has write rights on it as well as special permissions.



The `dnsNode` was created and our malicious record is in place, but there is a problem: with this default DACL the machine account of the brand new computer will be unable to create its own A record, specifically because the DACL only put the attacker in control of the node, and any other record type of that name will be subjected to the attacker's DACL, allowing only privileged users to create further records: if the target was to join the domain in these conditions no one would be able to reach it with DNS.

This is where our special permissions come into play, because conveniently we have both `WriteOwner` and `WriteDacl`.

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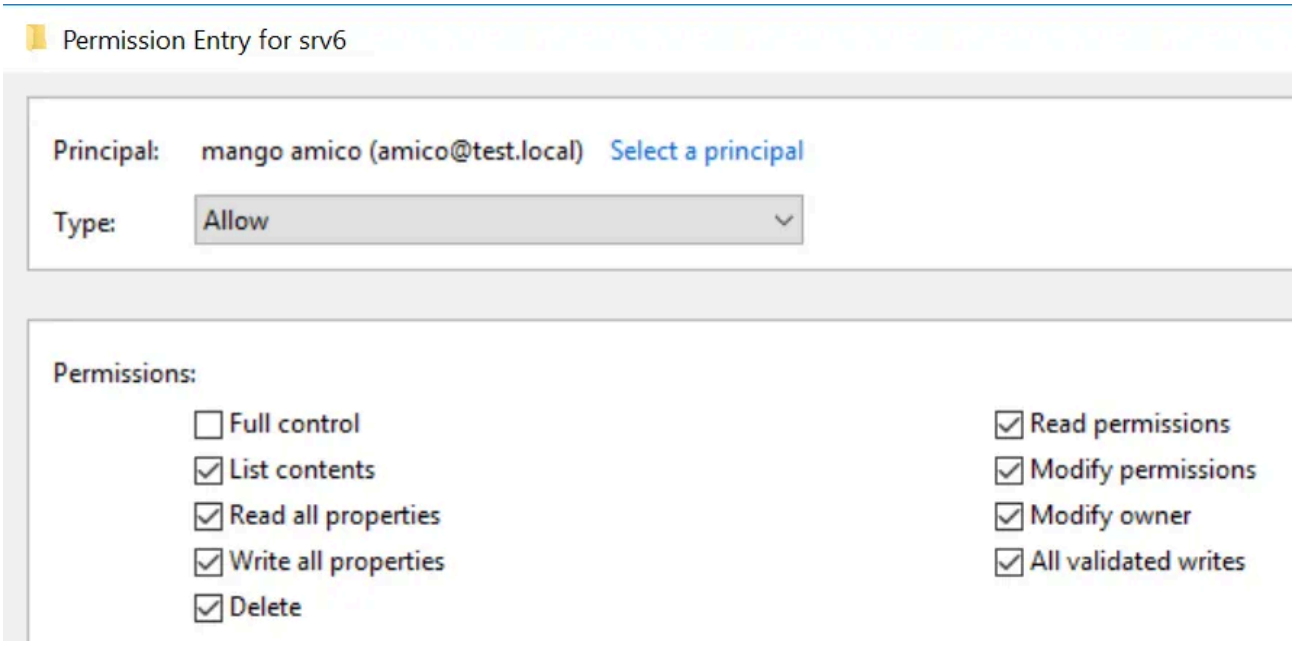

We can add an ACE giving other users write rights as well, this way the A record can be appended to our node's `dnsRecord` while retaining complete control over the node, and therefore over the new records too. We still need to be careful though, granting `GenericAll` or `GenericWrite` rights will actually break the attack.

After boot Windows will attempt to resolve its own hostname, if it does not receive an IP or if the address is outdated it will attempt to create a new record or replace the old one, using its machine account. If this account has full write rights on the record the node's DACL is replaced with a new one, where the machine account is the new owner and sole controller, getting rid of the attacker's ACE even though it was the original creator.

I found that to avoid this we can use `WriteProperty`, which limits modifications to only specific attributes. In my tests this is enough to allow domain hosts to create and consistently update their A record while keeping the attacker as the owner.

On Windows Powermad has a cmdlet to change a node's DACL:

```
Grant-ADIDNSPermission -Node SRV6 -Principal "Authenticated Users" -Access
WriteProperty
```

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Now the future machine account can create its record without us losing access.

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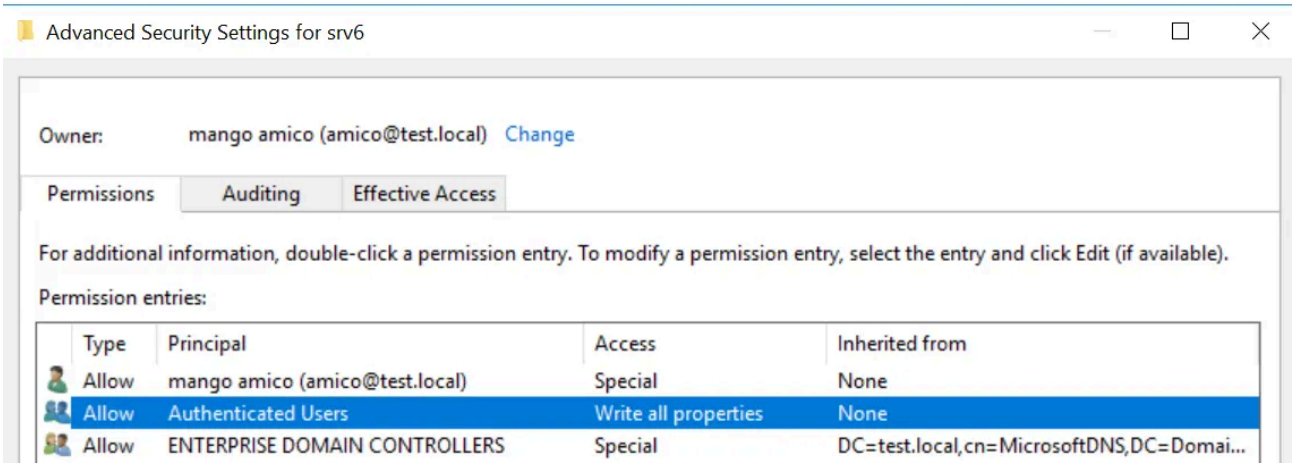

The main Linux tools with DACL edit features would not work with DNS records. Impacket's dacledit and bloody-AD do not support the `WriteProperty` right, while ldap_shell does but none of these three tools can even find the records, probably because they are stored in partitions that aren't automatically searched by default, like `DnsDomainZones`. In order to search these containers you need to set their DN as the search base, but the tools above don't allow it.

godap solves all these issues: it can explore and play with pretty much everything LDAP can do, we can browse the entire directory, create or modify objects and their DACL at will, and it even parses ADIDNS records and GPOs. Plus, it's written in Go so it runs on Windows too.

As I mentioned the containers of ADIDNS data aren't found by queries with the domain root as base DN, so we specify the right container to use as root:

```
godap 192.168.0.100 -u amico@test.local -p 'AAAAaaaa!1' -r
'DC=DomainDnsZones,DC=test,DC=local'
```

For some reason this tree view has trouble reading the names of some of the objects contained here, but you can still see the selected object's name from the Attributes panel.

Traverse the DNS hierarchy by expanding `MicrosoftDNS` and then the target zone name, where we find the exploitable record.

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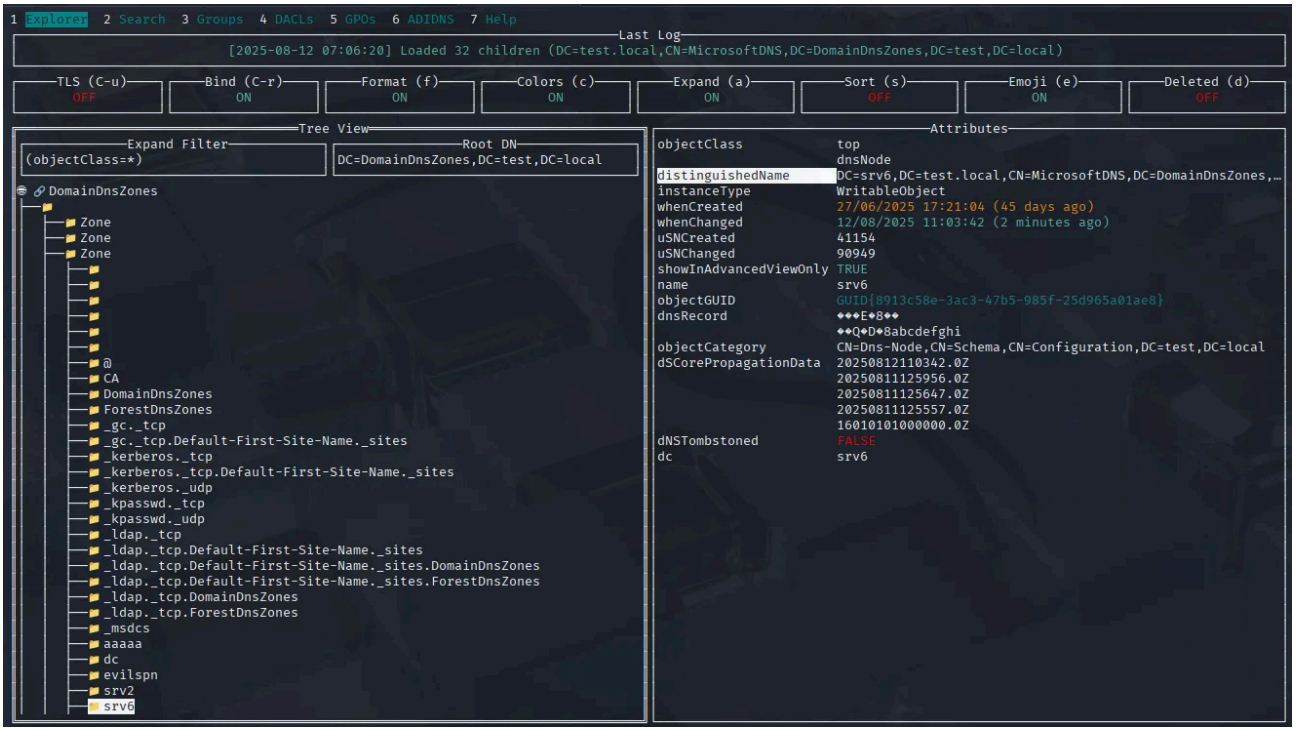

With the record selected we open its DACL with `Ctrl + D`, and then `Ctrl + N` to add a new ACE. Here we type the name of the principal we want to give access to, like Domain Computers, and press enter to let godap obtain the trustee's SID, which will appear on the bottom right. Select Allow as Type and double click on Write all properties, and Create.

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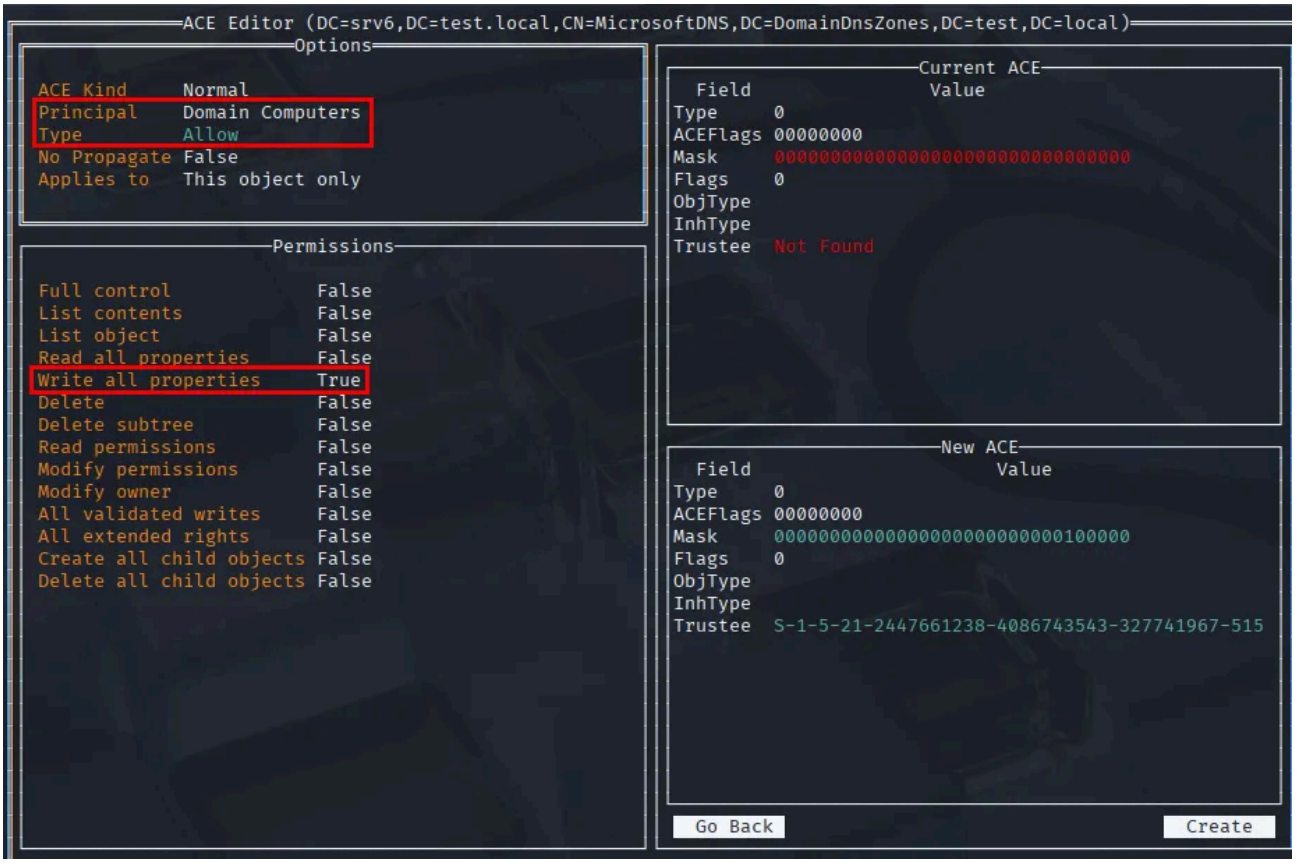

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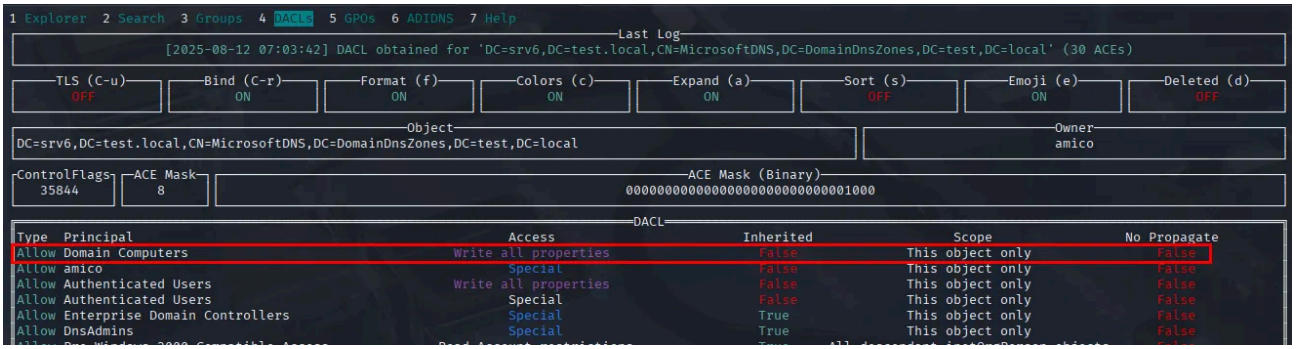
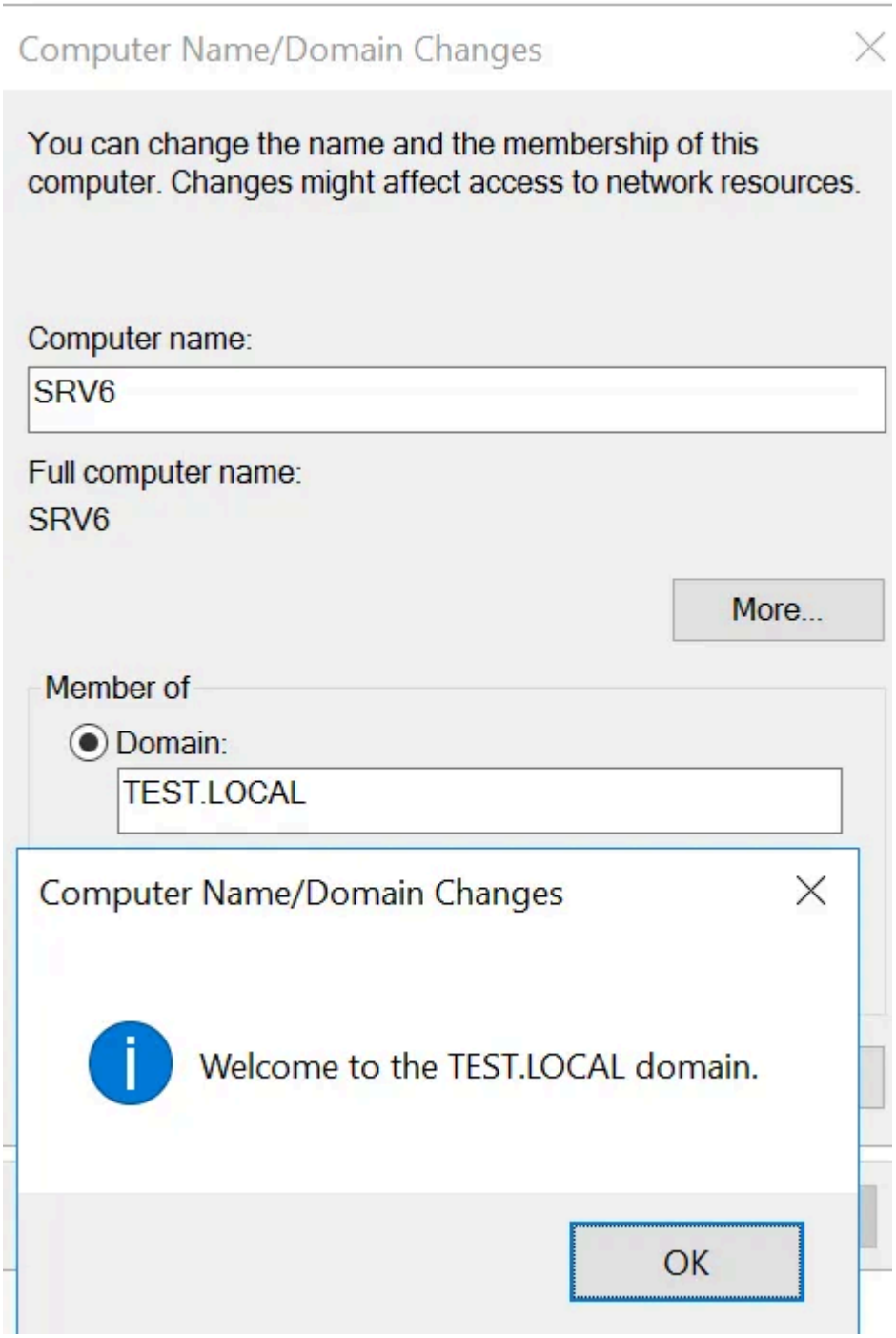


Since our target is a computer that still doesn't exist we can't add an ACE for it, we have to use a group we know it will be a part of so that it automatically inherits these privileges on creation. If we use a group like Domain Computers or Authenticated Users we should make sure to clean up the DACL at the end of the assessment, so that we do not leave exploitable DNS records up for grabs.

Now let's add a real computer to the domain with the name SRV6:



We see the A record is created successfully, and the DACL is identical to the TXT record we made ourselves:

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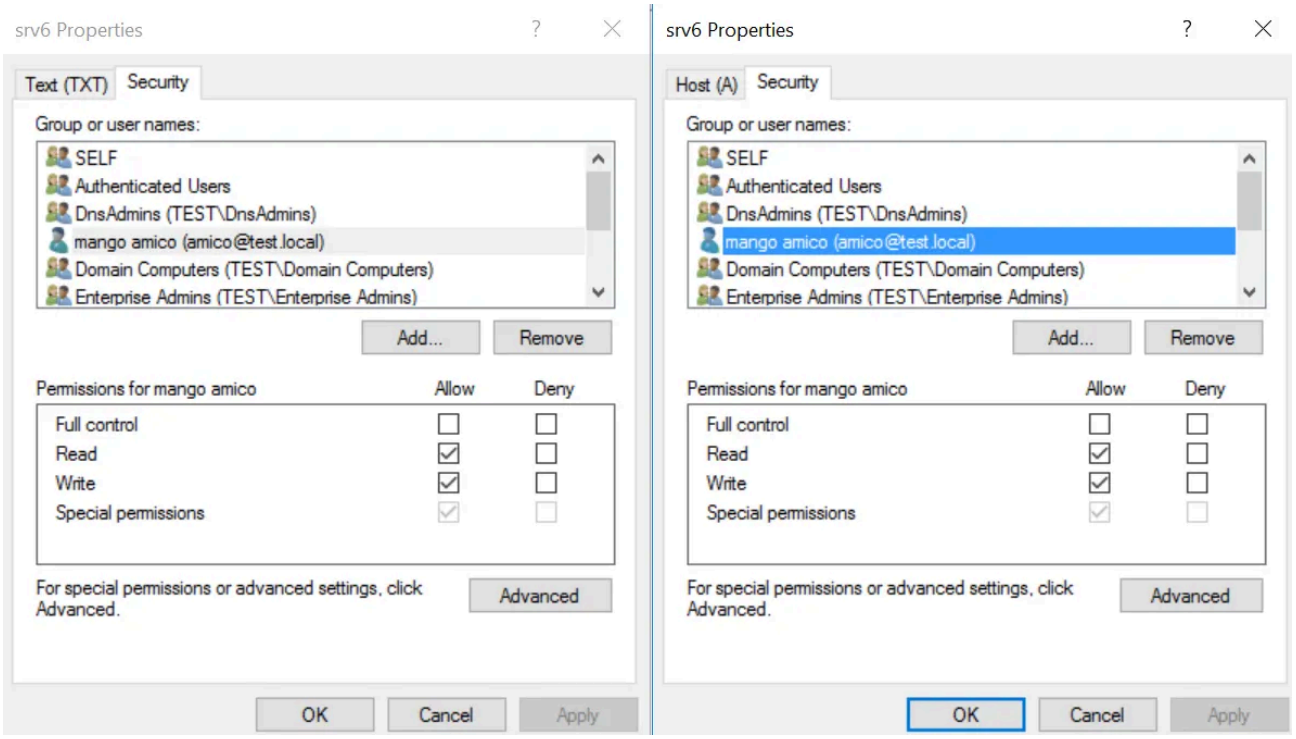

The node's `dnsRecord` now has two entries:

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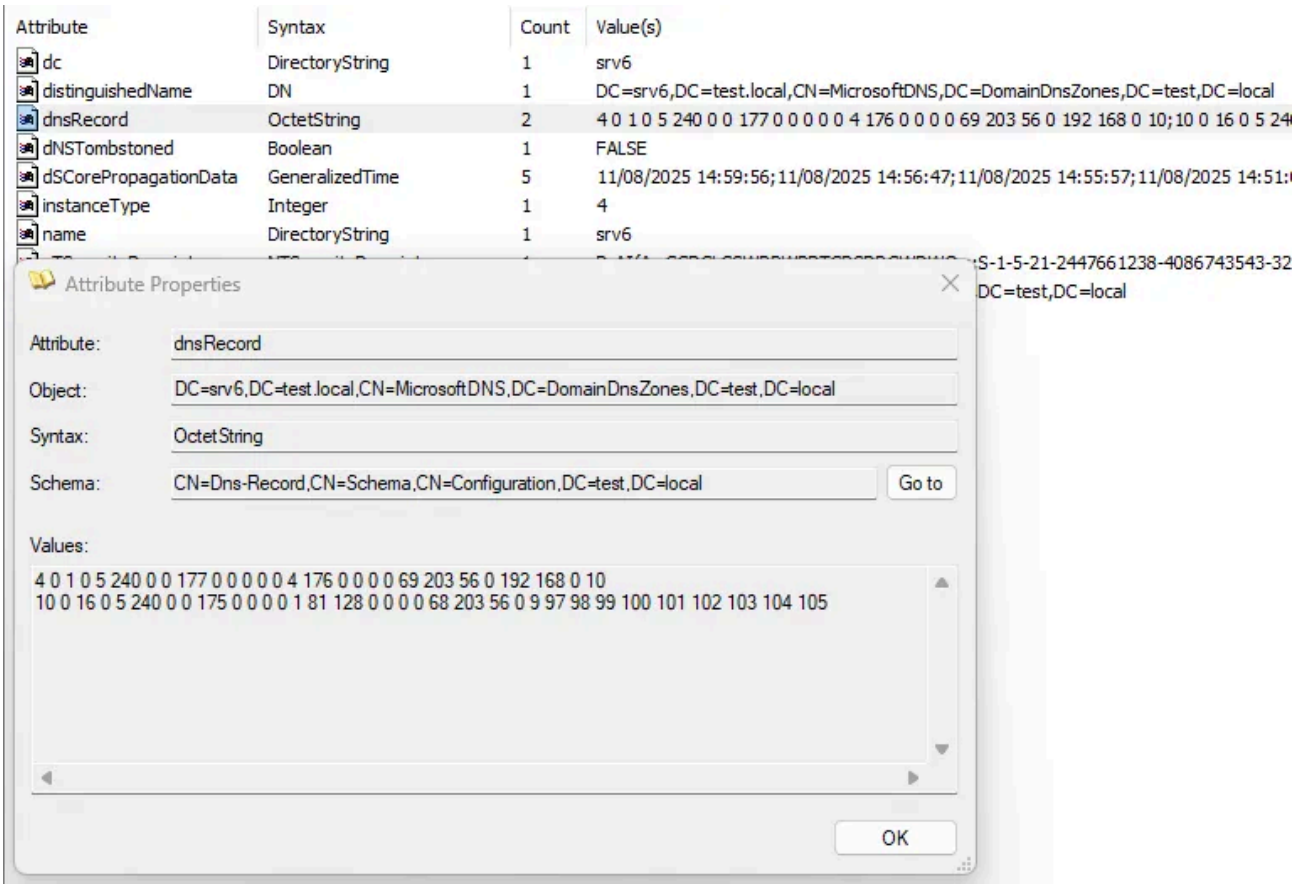

We Wait…

Once we have the malicious record set up we need to known when the computer is added to the domain so that we can launch the attack. The easiest way would probably be to regularly poll the `dnsNode`, monitoring its attributes for any changes. Alternatively we can use change notifications: LDAP clients can use asynchronous requests to receive instant notifications in the event of a change to a target DN, this system doesn't involve polling and should be more accurate because every change triggers a notification, with the client receiving a list of desired attributes with each notification. Of course this only works as long as the LDAP session is open.

In .NET these requests are implemented in System.DirectoryServices.Protocols, so for fun I used this example as a base for a dumb little tool that monitors a `dnsNode` for new A records, which you can find on my GitHub.

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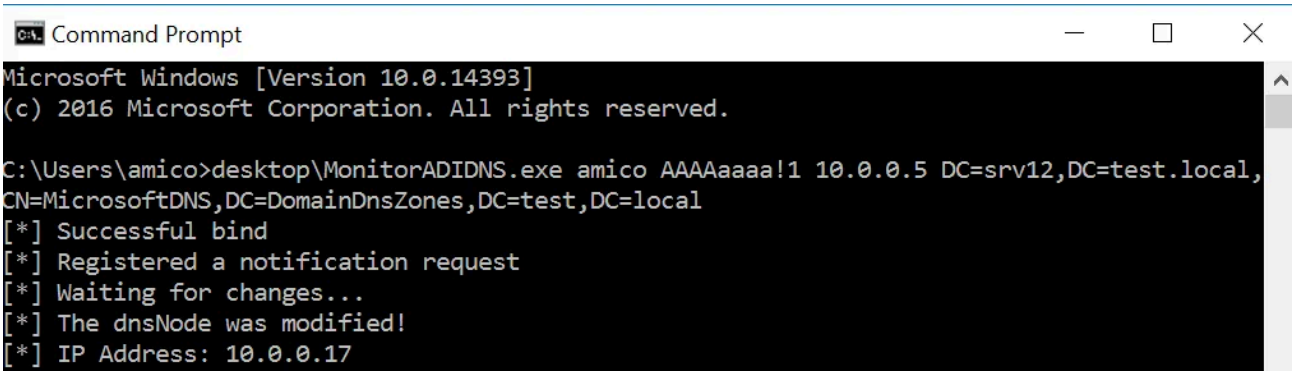

This is obviously a little experiment and not proper offensive tooling, I discovered a new feature and wanted to play with it. Change notification requests should also be part of python-ldap supported extended controls. Or, if you're not in a hurry, just run a query every now and then. In one way or the other, we gotta wait.

… And Relay!

Great, turns out we were right and a computer with the hostname we picked was actually joined to the domain! We can replace its legitimate IP with ours, causing every client connecting to SRV6 to reach out to us instead:

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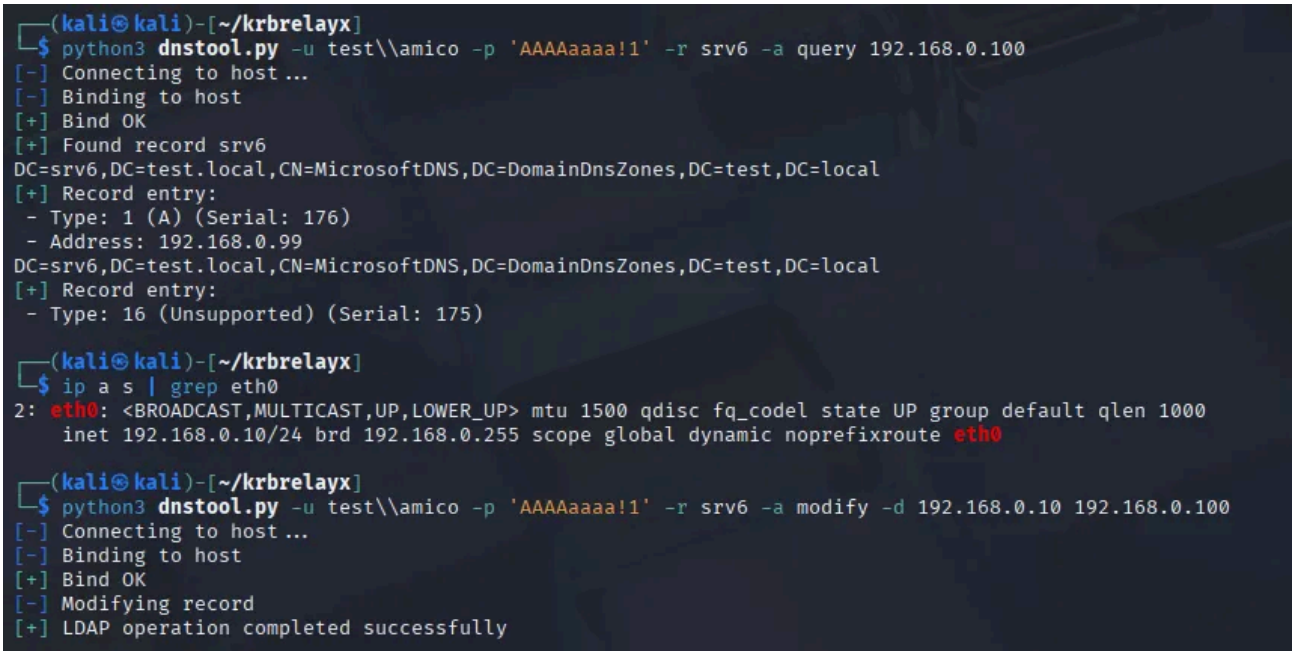

We are now in a MITM position and we control all the traffic sent to our vicitm. If signing isn't enforced this is the perfect occasion to attempt kerberos relays, especially if the host has just been added to the domain, we could reasonably expect some administrators to connect to it pretty soon to finalize the installation, allowing us complete control over the computer.

Right now the best tool for SMB AP-REQ relays is KrbRelayEx, it allows us to obtain a relay while also redirecting all legitimate connection attempts to the real host's IP so that clients can still work on the target during the attack, as without this step we would be causing a DoS. We can even run it on Linux after installing the .NET 8.0 runtime.

```
sudo dotnet KrbRelayEx.dll -spn cifs/srv6.test.local -secrets -redirecthost
192.168.0.99 -redirectports 5985,3389,135
```

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KrbRelayEx will intercept SMB authentication over kerberos, but a variant for relaying RPC authentication also exists, KrbRelayEx-RPC. Both tools are written by Andrea Pierini, who also wrote a great article to understand the basics of these relays.

Conclusion

This little trick should be one more reason to think about removing the default permissions allowing literally anyone to create arbitrary DNS records and enforcing signing, the only true defense against relays.

One consideration: when modifying existing A records you should keep in mind that machine accounts will eventually notice the modified A record and update it, so you should regularly check the value of the modified record to see if it was refreshed, and restore the record to its original IP when you're done.

ADIDNS is a quite fascinating target for AD attacks, so far research has mostly been focused around wildcard records but I am sure there is a lot more to uncover, this strange technique being just one example.